

Unit 04: Percentage

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Objective

The key concepts explained in this unit, including area a learner is expected to learn and master after going through the unit are: -

- Understand percentage commodity price increase/decrease
- Understand successive percent changes and budget based problems

Introduction

Concept of Percentage: By a certain percent, we mean that many hundredths. Thus x percent means x hundredths, written as x%. To express x% as a fraction: We have, $x\% = \frac{x}{100}$.

Thus, $20\% = \frac{20}{100} = \frac{1}{5}$; $48\% = \frac{48}{100} = \frac{12}{25}$, etc. To express a/b as a percent: We have, $\frac{a}{b} = \left(\frac{a}{b} \times 100\right)\%$. Thus, $\frac{1}{4} = \left[\left(\frac{1}{4}\right) \times 100\right] = 25\%$; $0.6 = \frac{6}{10} = \frac{3}{5} = \left[\left(\frac{3}{5}\right) \times 100\right]\% = 60\%$

The term percent means per hundred or for every hundred. It is the abbreviation of the Latin phrase *per centum*. Scoring 60 per cent marks means out of every 100 marks the candidate scored 60 marks. The term percent is sometimes abbreviated as p.c. The symbol % is often used for the term percent. Thus, 40 percent will be written as 40%. A fraction whose denominator is 100 is called a percentage and the numerator of the fraction is called rate percent, e.g., $\frac{5}{100}$ and 5 percent means the same thing, i.e., 5 parts out of every hundred parts.

4.1 Basic Formulae

01 To convert a fraction into a percent: To convert any fraction $\frac{1}{m}$ to rate percent, multiply it by 100 and put % sign, i.e., $\frac{1}{m} \times 100\%$



Example 1: What percentage is equivalent to $\frac{3}{5}$?

Solution: $\frac{3}{5} \times 100 = 60\%$.

02 To convert a per cent into a fraction: To convert a percent into a fraction, drop the percent sign and divide the number by 100.



Example 2: What fraction is $16\frac{2}{3}\%$?

$$\text{Solution: } 16\frac{2}{3}\% = \frac{\left(\frac{50}{3}\right)}{100} = \left(\frac{50}{3} \times \frac{1}{100}\right) = \frac{1}{6}$$

03 To find a percentage of a given number: $x\%$ of given number (N) = $\frac{x}{100} \times N$



Example 3: 75% of 400 =?

$$\text{Solution: } 75\% \text{ of } 400 = \frac{75}{100} \times 400 = 300.$$



Example 4: Find a number whose 4% is 72.

Solution: Let the required number be x. Then, 4% of x = 72

$$\Rightarrow \frac{4}{100} \times x = 72 \Rightarrow x = \frac{100}{4} \times 72 = 1800.$$



Example 5: What per cent of 25 Kg is 3.5 Kg?

Solution: Let $x\%$ of 25 Kg be 3.5 Kg.

$$\text{Then, } x\% \text{ of } 25 \text{ Kg} = 3.5 \text{ Kg} \Rightarrow x \times 100 \times 25 = 3.5$$

$$\Rightarrow x = \frac{x}{100} \times 25 = 3.5 \Rightarrow x = \frac{3.5 \times 100}{25} = 14.$$

Hence, 3.5 Kg is 14% of 25 Kg.

4.2 Short-cut Methods

01 (a) If A is $x\%$ more than that of B, then B is less than that of A by

$$\left[\frac{x}{100+x} \times 100 \right] \%$$

(b) If A is $x\%$ less than that of B, then B is more than that of A by

$$\left[\frac{x}{100-x} \times 100 \right] \%$$

Explanation

$$\text{Given } A = B + \frac{x}{100} B = \frac{100+x}{100} B$$

$$\therefore A - B = \frac{100+x}{100} B - B$$

$$= \left(\frac{100+x}{100} - 1 \right) B = \frac{x}{100} B$$

$$\text{So, } \frac{A-B}{A} = \frac{\frac{x}{100} B}{\frac{100+x}{100} B} = \frac{x}{100+x}$$

$$\Rightarrow A - B = \left(\frac{x}{100+x} \times 100 \right) \% \text{ of } A.$$

Therefore, B is less than that of A by

$$\left(\frac{x}{100+x} \times 100 \right) \%$$

Similarly, (b) can be proved.



Example 6: If Mohan's salary is 10% more than that of Sohan, then how much per cent is Sohan's salary less than that of Mohan?

Solution: Here, $x = 10$.

$\therefore$ Required answer =

$$\begin{aligned} & \left(\frac{x}{100+x} \times 100 \right) \% \\ & = \\ & \left(\frac{x}{100+x} \times 100 \right) \% \\ & = 9\frac{1}{11} \% \end{aligned}$$



Example 7: If A's income is 40% less than B's income, then how much per cent is B's income more than A's income?

Solution: Here, $x = 40$

$$\begin{aligned} \therefore \text{Required} & \qquad \qquad \qquad \text{answer} & = \\ = & \left(\frac{x}{100-x} \times 100 \right) \% \\ & = \left(\frac{40}{100-40} \times 100 \right) \% = 66\frac{2}{3} \% \end{aligned}$$

02 If A is $x\%$ of C and B is $y\%$ of C, then $A = \frac{x}{y} \times 100\%$ of B

Explanation

$$\text{Given } A = \frac{x}{100} C \Rightarrow C = 100 \frac{A}{x}$$

$$\text{And, } B = \frac{y}{100} C \Rightarrow C = 100 \frac{B}{y}$$

$$\therefore C = 100 \frac{A}{x} = 100 \frac{B}{y} \Rightarrow A = \frac{x}{y} B$$

$$\text{Or, } \frac{x}{y} \times 100\% \text{ of } B$$



Example 8: If A is 20% of C and B is 25% of C, then what percentage is A of B?

Solution: Here, $x = 20$ and $y = 25$.

$$A = \frac{x}{y} \times 100\% \text{ of } B$$

$$= \frac{20}{25} \times 100\% \text{ of } B, \text{ i.e., } 80\% \text{ of } B$$

03 (a) If two numbers are respectively $x\%$ and $y\%$ more than a third number, then the first number is $\left(\frac{100+x}{100+y} \times 100 \right) \%$ of the second and the second number is $\left(\frac{100-y}{100-x} \times 100 \right) \%$ of the first

(b) If two numbers are, respectively, $x\%$ and $y\%$ less than a third number, then the first number is $\left(\frac{100-x}{100-y} \times 100 \right) \%$ of the second

and the second number is $\left(\frac{100-y}{100-x} \times 100 \right) \%$

Explanation Let A, B and C be the three numbers. Given:

$$A = C + \frac{x}{100} C = \left(\frac{100+x}{100} \right) C \Rightarrow C = A \left(\frac{100}{100+x} \right)$$

$$\text{And, } B = C + \frac{y}{100} C = \left(\frac{100+y}{100} \right) C \Rightarrow C = B \left(\frac{100}{100+y} \right)$$

$$\therefore A \left(\frac{100}{100+x} \right) = B \left(\frac{100}{100+y} \right)$$

$$\Rightarrow A = \left(\frac{100+y}{100+x} \right) B \text{ Or } \left(\frac{100+y}{100+x} \right) \times 100\% \text{ of } B$$

$$\text{and, } B = \left(\frac{100+x}{100+y} \right) A \text{ or } \left(\frac{100+x}{100+y} \right) \times 100\% \text{ of } A.$$

Similarly, (b) can be proved



Example 9: Two numbers are respectively 20% and 50% more than a third number. What per cent is the first of the second?

Solution: Here, $x = 20$ and $y = 50$.

$$\therefore \text{First number} = \left(\frac{100+x}{100+y} \right) \times 100\% \text{ of the second}$$

$$= \left(\frac{100+20}{100+50} \right) \times 100\% \text{ of the second}$$

i.e., 80% of the second.



Example 10: Two numbers are, respectively, 32% and 20% less than a third number. What per cent is the first of the second?

Solution: Here, $x = 32$ and $y = 20$.

$$\therefore \text{First number} = \left(\frac{100-x}{100-y} \right) \times 100\% \text{ of the second}$$

$$= \left(\frac{100-32}{100-20} \right) \times 100\% \text{ of the second}$$

i.e., 85% of the second.

04 (a) If the price of a commodity increases by $P\%$, then the reduction in consumption so as not to increase the expenditure is

$$\left(\frac{p}{100+p} \times 100 \right) \%$$

(b) If the price of a commodity decreases by $P\%$, then the increase in consumption so as not to decrease the expenditure is

$$\left(\frac{p}{100-p} \times 100 \right) \%$$

Explanation Let the original price of the commodity be rs.100. Then, the increased price = $100 - \frac{p}{100} \times 100$

$$= \text{rs } (100 + P).$$

Therefore, to keep the price unchanged, there should be a reduction in the consumption of the commodity by rs.P.

$$\text{Decrease in rs } (100 + P) = \text{rs. } P$$

$$\therefore \text{Decrease in rs } 100 =$$

$$\frac{p}{100+p} \times 100$$

$\therefore$ Required reduction in consumption is

$$\left(\frac{p}{100+p} \times 100 \right) \%$$

Similarly, (b) part can be proved.



Example 11: If the price of sugar increases by 25%, find how much per cent its consumption be reduced so as not to increase the expenditure.

Solution: Reduction in consumption

$$\left(\frac{p}{100+p} \times 100 \right) \%$$

=

$$\left(\frac{25}{100+25} \times 100 \right) \% \text{ or } 20\%$$



Example 12: If the price of a commodity decreases by 25%, find how much per cent its consumption be increased so as not to decrease the expenditure.

Solution: Increase in consumption

$$= \left(\frac{p}{100 - p} \times 100 \right) \%$$

$$= \left(\frac{25}{100 - 25} \times 100 \right) \% \text{ or } 33\frac{1}{3} \%$$

05 If a number is changed (increased/decreased) successively by $x\%$ and $y\%$, then net % change is given by $\left(x + y + \frac{xy}{100}\right) \%$ which represents increase or decrease in value according as the sign is +ve or -ve. If x or y indicates decrease in percentage, then put -ve sign before x or y , otherwise +ve sign.

Explanation

Let the given number be N .

If it is increased by $x\%$, then it becomes

$$N + x\% \text{ of } N = N + \frac{Nx}{100} = \frac{N(x+100)}{100}$$

If it is further increased by $y\%$, then it becomes

$$\frac{N(x+100)}{100} + \frac{y}{100} \times \frac{N(x+100)}{100}$$

$$= \frac{N(x+100)(y+100)}{(100)^2}$$

$$\therefore \text{Net change} = \frac{N(x+100)(y+100)}{(100)^2}$$

$$= \frac{N(100x + 100y + xy)}{(100)^2}$$

$$\therefore \% \text{ change} = N \left(x + y + \frac{xy}{100} \right) \times \frac{1}{100} \times \frac{100}{N}$$

$$\left(x + y + \frac{xy}{100} \right) \%$$



Example 13: If salary of a person is first increased by 15% and thereafter decreased by 12%, what is the net change in his salary?

Solution: Here, $x = 15$ and $y = -12$.

$\therefore$ The net % change in the salary

$$= \left(x + y + \frac{xy}{100} \right) \% = \left(15 - 12 - \frac{15 \times 12}{100} \right) \% \text{ or } 1.2\%$$

Since the sign is +ve, the salary of the person increases by 1.2%



Example 14: The population of a town is decreased by 25% and 40% in two successive years. What per cent population is decreased after two years?

Solution: Here, $x = -25$ and $y = -40$.

$\therefore$ The net % change in population

$$= \left(x + y + \frac{xy}{100} \right) \%$$

$$= \left(-25 - 40 + \frac{25 \times 40}{100} \right) \% \text{ or } -55\%$$

Analytical Skills-I

Since the sign is -ve, there is decrease in population after two years by 55%

06 If two parameters A and B are multiplied to get a product and if A is changed (increased/decreased) by x% and another parameter B is changed (increased/decreased) by y%, then the net % change in the product ($A \times B$) is given

$\left(x + y + \frac{xy}{100}\right) 100\%$ which represents increase or decrease in value according as the sign in +ve or -ve. If x or y indicates decrease in percentage, then put -ve sign before x or y, otherwise +ve sign. \



Example 15: If the side of a square is increased by 20%, its area is increased by k%. Find the value of k.

Solution: Since side $\times$ side = area

$\therefore$ Net% change in area

$$\left(x + y + \frac{xy}{100}\right) \% = \left(20 + 20 + \frac{20 \times 20}{100}\right) \%$$

=44%

Therefore, the area is increased by 44%. Here, k = 44.



Example 16: The radius of a circle is increased by 2%. Find the percentage increase in its area.

Solution: Since $\pi \times \text{radius} \times \text{radius} = \text{area} \therefore$ Net% change in area

$$= \left(x + y + \frac{xy}{100}\right) \% = \left(2 + 2 + \frac{2 \times 2}{100}\right) \%$$

[Here, x = 2 and y = 2]

$$= 4\frac{1}{25} \%$$

Therefore, the percentage increase in area is $4\frac{1}{25} \%$



Example 17: The tax on a commodity is diminished by 15% and its consumption increases by 10%. Find the effect on revenue.

Solution: Since tax $\times$ consumption = revenue $\therefore$ Net% change in revenue

$$= \left(x + y + \frac{xy}{100}\right) \% = \left(-15 + 10 + \frac{15 \times 10}{100}\right) \%$$

[Here, x = -15 and y = 10]

$$= -6.5\%$$

$\therefore$ The revenue decreases by 6.5%

07 If the present population of a town (or value of an item) be P and the population (or value of item) changes at r% per annum, then

(a) Population (or value of item) after n years

$$= p \left(1 + \frac{r}{100}\right)^n$$

(b) Population (or value of item) n years ago

$$= \frac{p}{\left(1 + \frac{r}{100}\right)^n}$$

here r is +ve or -ve according as the population

(or value of item) increases or decreases.

Explanation

Population at the end of first year

$$=P + \frac{r}{100}P = P\left(1 + \frac{r}{100}\right)$$

Now, the population at the beginning of second year

$$=P\left(1 + \frac{r}{100}\right)$$

∴ Population at the end of second year

$$=P\left(1 + \frac{r}{100}\right) + \frac{r}{100}P\left(1 + \frac{r}{100}\right) = P\left(1 + \frac{r}{100}\right)^2$$

$$\text{Population at the end of } n \text{ years} = P\left(1 + \frac{r}{100}\right)^n$$



Example 18: The population of a town increases 5% annually. If its present population is 84000, what will it be in 2 years time?

Solution: Here, $P = 84000$, $r = 5$ and $n = 2$.

∴ Population of the town after 2 years

$$=P\left(1 + \frac{r}{100}\right)^n = 84000P\left(1 + \frac{5}{100}\right)^2$$

$$=84000 \times \frac{105}{100} \times \frac{105}{100} = 92610$$



Example 19: The population of a town increases at the rate of 5% annually. If the present population is 4410, what it was 2 years ago?

Solution: Here, $P = 4410$, $r = 5$ and $n = 2$.

∴ Population of the town 2 years ago

$$= \frac{P}{\left(1 + \frac{r}{100}\right)^n} = \frac{4410}{\left(1 + \frac{5}{100}\right)^2} = \frac{4410}{\frac{105}{100} \times \frac{105}{100}} = 4000$$

08 If a number A is increased successively by $x\%$ followed by $y\%$ and then by $z\%$, then the final value of A will be

$$A\left(1 + \frac{x}{100}\right)\left(1 + \frac{y}{100}\right)\left(1 + \frac{z}{100}\right)$$

In case a given value decreases by any percentage, we will use a -ve sign before that.



Example 20: The population of a town is 144000. It increases by 5% during the first year. During the second year, it decreases by 10% and increases by 15% during the third year. What is the population after 3 years?

Solution: Here, $P = 144000$, $x = 5$, $y = -10$ and $z = 15$.

∴ Population of the town after 3 years

$$=A\left(1 + \frac{x}{100}\right)\left(1 + \frac{y}{100}\right)\left(1 + \frac{z}{100}\right)$$

$$=144000\left(1 + \frac{5}{100}\right)\left(1 - \frac{10}{100}\right)\left(1 + \frac{15}{100}\right)$$

$$= \frac{144000 \times 105 \times 90 \times 115}{1001 \times 1001 \times 100} = 156492$$

Analytical Skills-I

09 In an examination, the minimum pass percentage is $x\%$. If a student secures y marks and fails by z marks, then the maximum marks in the examination is $\frac{100(y+z)}{x}$.

Explanation Let the maximum marks be m . Given: $x\%$ of $m = y + z$

$$\Rightarrow \frac{x}{100} \times m = y + z \text{ or } m = \frac{100(y+z)}{x}$$



Example 21: In an examination, a student must get 60% marks to pass. If a student who gets 120 marks, fails by 60 marks, find the maximum marks.

Solution: Here, $x = 60$, $y = 120$ and $z = 60$.

$\therefore$ Maximum marks

$$= \frac{100(y+z)}{x} = \frac{100(120+60)}{60} = \frac{100 \times 180}{60} = 300$$

10 In an examination $x\%$ and $y\%$ students respectively fail in two different subjects while $z\%$ students fail in both the subjects, then the percentage of students who pass in both the subjects will be $(100 - (x + y - z))\%$

Explanation

Percentage of students who failed in one subject = $(x - z)\%$

Percentage of students who failed in other subject = $(y - z)\%$

Percentage of students who failed in both the subjects = $z\%$

$\therefore$ Percentage of students who passed in both the subjects = $[100 - [(x - z) + (y - z) + z]]\% = (100 - (x + y - z))\%$



Example 22: In an examination, 42% students failed in Mathematics and 52% failed in Science. If 17% failed in both the subjects, find the percentage of those who passed in both the subjects.

Solution: Here, $x = 42$, $y = 52$ and $z = 17$.

$\therefore$ Percentage of students passing both the subjects = $(100 - (x + y - z))\% = (100 - (42 + 52 - 17))\%$ or 23%.

In this article, we deal with the concept of successive percentage change. This is a problem type in Percentages and using the formula in this article, you can easily solve questions based on this concept in matter of seconds.

4.3 Successive Percentage Change

What is Successive Percentage Change?

The concept of successive percentage change deals with two or more percentage changes applied to quantity consecutively. In this case, the final change is not the simple addition of the two percentage changes (as the base changes after the first change).

Formula for Percentage Change:

Suppose a number N undergoes a percentage change of $x\%$ and then $y\%$, the net change is:

$$\text{New number} = N \times (1 + x/100) \times (1 + y/100).$$

$$\text{Now, } (1 + x/100) \times (1 + y/100) = 1 + x/100 + y/100 + xy/10000$$

$$\text{If we say that } x + y + xy/100 = z, \text{ then } (1 + x/100) \times (1 + y/100) = 1 + z/100$$

Here, z is the effective percentage change when a number is changed successively by two percentage changes.

Various cases for Percentage Change:

Both percentage changes are positive:

x and y are positive and net increase = $(x+y+xy/100)$ %.

One percentage change is positive and the other is negative:

x is positive and y is negative, then net percentage change = $(x-y-xy/100)$ %

Both percentage changes are negative:

x and y both are negative and imply a clear decrease = $(-x-y+xy/100)$ %

Percentage Change involving three changes:

If value of an object/number P is successively changed by $x\%$, $y\%$ and then $z\%$, then final value.

$$=P(1 \pm \frac{x}{100})(1 \pm \frac{y}{100})(1 \pm \frac{z}{100})$$



Example 23: A's salary is increased by 10% and then decreased by 10%. The change in salary is

Solution:

Percentage change formula when x is positive and y is negative = $\{x - y - (xy/100)\}$ %

Here, $x = 10$, $y = 10$

$$= \{10 - 10 - (10 \times 10)/100\} = -1\%$$

As negative sign shows a decrease, hence the final salary is decreased by 1%.

4.4 Weighted Averages and Expected Values

- Find weighted averages, and compute and interpret expected values. Visitors to a Web site give books ratings from 1 to 5 stars. The chart shows the ratings that one book received from 10 visitors to the site. To find the average rating, you can add the numbers of stars in the ratings and then divide by the number of ratings.
 $[4 + 1 + 3 + 5 + 3 + 5 + 2 + 5 + 5 + 4]/10 = 3.7$



- The average rating is 3.7 stars. You can also find the average rating by multiplying each rating by the number of times it occurs, adding these products, and then dividing by the total number of ratings.
 $[1(1) + 2(1) + 3(2) + 4(2) + 5(4)]/10 = 3.7$
- Notice that some ratings occur more frequently than others, and these ratings are "weighted" more heavily in the numerator because they are multiplied by greater values. This idea can be generalized to give a definition of a weighted average. Note that the weights do not have to be based on frequencies, but can be based on other factors, such as relative importance or value.

Summary

The key concepts learned from this unit are: -

- We have learnt percentage commodity price increase/ decrease
- We have learnt successive percent changes and budget based problems.

Keywords

- Percentage
- Commodity
- Successive percent
- Budget

Self Assessment

1. Express 56% as a fraction:

- A. (a)14/25
- B. (b)18/25
- C. (c)19/25
- D. (d)20/25

2. Express 4% as a fraction:

- A. 1/24.
- B. 1/20.
- C. 3/25.
- D. 1/25.

3. Express 0.6% as a fraction:

- A. 3/70
- B. 3/125
- C. 13/50
- D. 3/500

4. Express 0.008% as a fraction:

- A. 12/1250
- B. 13/1250
- C. 1/1250
- D. 7/1250

5. Express 6% as a Decimal:

- A. 0.06.
- B. 0.28.
- C. 0.002.
- D. 0.004

6. Express 28% as a Decimal:

- A. 0.06.

- B. 0.28.
C. 0.002.
D. 0.004
7. Express 0.04% as a Decimal :
A. 0.06.
B. 0.28.
C. 0.002.
D. 0.004
8. 2 is what percent of 50 ?
A. 4%
B. 12%
C. 0.2%
D. 0.04%
9. ?% of 25 = 20125.
A. 12.5
B. 9
C. 8.5
D. 14.5
10. Which is greatest in $16\frac{2}{3}\%$, $\frac{2}{5}$ and 0.17 ?
A. $\frac{2}{5}$
B. 16
C. $\frac{2}{3}$
D. 0.17
11. An inspector rejects 0.08% of the meters as defective. How many will be examined to project ?
A. 2400
B. 2500
C. 3000
D. 4000
12. Sixty five percent of a number is 21 less than four fifth of that number. What is the number ?
A. 130
B. 140
C. 150
D. 200
13. In expressing a length 810472 km as nearly as possible with three significant digits, find the percentage error
A. 0.0034
B. 0.034
C. 0.34
D. 3.4

Analytical Skills-I

14. if 50% of $(x-y) = 30\%$ of $(x+y)$ then what percent of x is y ?
- A. 25%
- B. 50%
- C. 75%
- D. 100%
15. The salary of a person was reduced by 10% .By what percent should hisreduced salary be raised so as to bring it at par with his original salary ?
- A. $(100/9)\%$
- B. $(20/9)\%$
- C. $(100/19)\%$
- D. $(100/29)\%$

Answers for Self Assessment

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. A | 2. D | 3. D | 4. C | 5. A |
| 6. B | 7. D | 8. A | 9. C | 10. D |
| 11. B | 12. B | 13. B | 14. A | 15. A |

Review Questions

- When the price for a product was decreased by 10%, the number sold increased by 30%. What was the effect on the total revenue?
- If the numerator of a fraction be increased by 15% and its denominator be diminished by 8% , the value of the fraction is $15/16$. Find the original fraction.
- The population of a town is 1, 76,400 . If it increases at the rate of 5% per annum, what will be its population 2 years hence? What was it 2 years ago?
- In the new budget, the price of kerosene oil rose by 25%. By how much percent must a person reduce his consumption so that his expenditure on it does not increase?
- The value of a machine depreciates at the rate of 10% per annum. If its present is Rs.1,62,000 what will be its worth after 2 years ? What was the value of the machine 2 years ago?
- During one year, the population of town increased by 5%. If the total population is 9975 at the end of the second year, then what was the population size in the beginning of the first year ?
- If A earns $99/3\%$ more than B, how much percent does B earn less than A?
- How many kg of pure salt must be added to 30kg of 2% solution of salt and water to increase it to 10% solution?
- In an examination, 80% of the students passed in English , 85% in Mathematics and 75% in both English and Mathematics. If 40 students failed in both the subjects, find the total number of students.
- The salary of a person was reduced by 20% .By what percent should his reduced salary be raised so as to bring it at par with his original salary ?

11. The capacity of a ground was 100000 at the end of 2012. In 2013, it increased by 10% and in 2014, it decreased by 18.18%. What was the ground's capacity at the end of 2014?

**Further Readings**

1. Quantitative Aptitude For Competitive Examinations By Dr. R S Aggarwal, S Chand Publishing
2. A Modern Approach To Verbal & Non-Verbal Reasoning By Dr. R S Aggarwal, S Chand Publishing
3. Magical Book On Quicker Maths By M Tyra, Banking Service Chronicle
4. Analytical Reasoning By M.K. Pandey, Banking Service Chronicle